

ESHAAN NICHANI

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EDUCATION

Princeton University • Princeton, NJ September 2021 – Present
PhD Candidate in Electrical & Computer Engineering
Advisors: Jason D. Lee, Yuxin Chen
GPA: 4.0/4.0

Massachusetts Institute of Technology • Cambridge, MA June 2020 – May 2021
Masters of Engineering in EECS
Thesis: *An Empirical and Theoretical Analysis of the Role of Depth in Convolutional Neural Networks*.
Advisor: Caroline Uhler

Massachusetts Institute of Technology • Cambridge, MA September 2016 – May 2020
Bachelors of Science in Mathematics
Bachelors of Science in Computer Science and Engineering
GPA: 5.0/5.0

RESEARCH

(* denotes equal contribution or alphabetical authorship)

Publications:

- **Quantitative Bounds for Length Generalization in Transformers.**
Zachary Izzo*, **Eshaan Nichani***, Jason D. Lee.
International Conference on Learning Representations (ICLR), 2026 (Oral).
- **Emergence and Scaling Laws for SGD Learning of Shallow Neural Networks.**
Yunwei Ren*, **Eshaan Nichani***, Denny Wu, Jason D. Lee
Advances in Neural Information Processing Systems (NeurIPS), 2025
- **Learning Compositional Functions with Transformers from Easy-to-Hard Data.**
Zixuan Wang*, **Eshaan Nichani***, Alberto Bietti, Alex Damian, Daniel Hsu, Jason D. Lee, Denny Wu
Conference on Learning Theory (COLT), 2025
- **Understanding Factual Recall in Transformers Via Associative Memories.**
Eshaan Nichani, Jason D. Lee, Alberto Bietti
International Conference on Learning Representations (ICLR), 2025 (Spotlight).
Mathematics of Modern Machine Learning (M3L) Workshop, NeurIPS, 2024 (Oral).
- **Learning Hierarchical Polynomials of Multiple Nonlinear Features with Three-Layer Networks.**
Hengyu Fu, Zihao Wang, **Eshaan Nichani**, Jason D. Lee
International Conference on Learning Representations (ICLR), 2025.
- **How Transformers Learn Causal Structure with Gradient Descent.**
Eshaan Nichani, Alex Damian, Jason D. Lee.
International Conference on Machine Learning (ICML), 2024.
- **Learning Hierarchical Polynomials with Three-Layer Neural Networks.**
Zihao Wang, **Eshaan Nichani**, Jason D. Lee.
International Conference on Learning Representations (ICLR), 2024.
- **Metastable Mixing of Markov Chains: Efficiently Sampling Low Temperature Exponential Random Graphs.**
Guy Bresler*, Dheeraj Nagaraj*, **Eshaan Nichani***.
Annals of Applied Probability, 2024.
- **Provable Guarantees for Nonlinear Feature Learning in Three-Layer Neural Networks.**
Eshaan Nichani, Alex Damian, Jason D. Lee.
Advances in Neural Information Processing Systems (NeurIPS), 2023 (Spotlight).
- **Smoothing the Landscape Boosts the Signal for SGD: Optimal Sample Complexity for Learning Single Index Models.**
Alex Damian, **Eshaan Nichani**, Rong Ge, Jason D. Lee.
Advances in Neural Information Processing Systems (NeurIPS), 2023 (Oral).
- **Fine-Tuning Language Models with Just Forward Passes.**
Sadhika Malladi, Tianyu Gao, **Eshaan Nichani**, Alex Damian, Jason D. Lee, Danqi Chen, Sanjeev Arora.
Advances in Neural Information Processing Systems (NeurIPS), 2023 (Oral).

- **Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability.**
Alex Damian*, **Eshaan Nichani***, Jason D. Lee.
International Conference on Learning Representations (ICLR), 2023.
- **Identifying good directions to escape the NTK regime and efficiently learn low-degree plus sparse polynomials.**
Eshaan Nichani, Yu Bai, Jason D. Lee.
Advances in Neural Information Processing Systems (NeurIPS), 2022.
- **Causal Structure Discovery between Clusters of Nodes Induced by Latent Factors.**
Chandler Squires, Annie Yun, **Eshaan Nichani**, Raj Agrawal, Caroline Uhler.
Conference on Causal Learning and Reasoning (CLear), 2022.
- **Assessment of circulant copy number variant detection for cancer screening**
Bhuvan Molparia, **Eshaan Nichani**, Ali Torkamani.
PLoS ONE, 2017.

Preprints:

- **Sharp Capacity Scaling of Spectral Optimizers in Learning Associative Memory.**
Juno Kim*, **Eshaan Nichani***, Denny Wu, Alberto Bietti, Jason D. Lee.
ArXiv preprint, 2026.
- **On the Statistical Query Complexity of Learning Semiautomata: a Random Walk Approach.**
George Giapitzakis, Kimon Fountoulakis, **Eshaan Nichani**, Jason D. Lee.
ArXiv preprint, 2025.

Workshop Papers:

- **Increasing Depth Leads to U-Shaped Test Risk in Over-parameterized Convolutional Networks.**
Eshaan Nichani*, Adityanarayanan Radhakrishnan*, Caroline Uhler.
Overparameterization: Pitfalls and Opportunities, ICML, 2021.
- **On Alignment in Deep Linear Neural Networks.**
Adityanarayanan Radhakrishnan*, **Eshaan Nichani***, Daniel Bernstein, Caroline Uhler.
Overparameterization: Pitfalls and Opportunities, ICML, 2021.
- **Adaptive diagonal curvature: a quasi-newton method for stochastic optimization**
David Saxton, **Eshaan Nichani**.
Beyond First Order Methods in Machine Learning Systems, ICML, 2020.

HONORS AND AWARDS

• Rising Star in Data Science (Stanford, UChicago, UC San Diego)	2025
• IBM PhD Fellowship	2024 – 2025
• Finalist for Two-Sigma PhD Fellowship	2024
• DoD National Defense Science and Engineering Graduate Fellowship	2021 – 2024
• Putnam Mathematics Competition – Rank N2 (top 25 in US)	2016, 2019
• Romanian Master of Mathematics International Competition – 1st place individual worldwide (Gold medal)	2016
• USA Mathematical Olympiad – Honorable Mention	2016

INVITED TALKS

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- *Foundations of Language Models: Scaling and Reasoning*
 - Computer Science Seminar, Columbia University, March 2026.
 - ECE RISE-AI Seminar Series, University of Wisconsin-Madison, February 2026.
 - ECE Seminar, University of Texas at Austin, February 2026.
 - Econometrics & Statistics Colloquium, UChicago Booth, January 2026.
 - Data-Driven Seminar, Stanford University, January 2026.
 - *Emergence and Scaling Laws for SGD Learning of Shallow Neural Networks*
 - Math and Data (MaD) Seminar, NYU, September 2025.
 - *Learning Compositional Functions with Transformers from Easy-to-Hard Data*
 - Flatiron Institute, New York, September 2025.
 - Formal Languages and Neural Networks (FLaNN) seminar, virtual, September 2025.
 - Statistical Physics & Machine Learning: Moving Forward, Cargese, August 2025
 - Conference on Learning Theory, Lyon, July 2025.
 - Jason Altschuler group meeting, UPenn, April 2025.
 - *Understanding Factual Recall in Transformers via Associative Memories*
 - Oral presentation, Mathematics of Modern Machine Learning (M3L) workshop, Vancouver, December 2024.

- *How Transformers Learn Causal Structure with Gradient Descent.*
 - Statistics Seminar, UC Davis, November 2025.
 - Alg-ML Seminar, Princeton University, November 2024.
 - INFORMS Annual Meeting, Seattle, October 2024.
 - Citadel PhD Colloquium, New York, May 2024.
 - Flatiron Institute, New York, March 2024.
 - Siemens Research, Princeton, February 2024.
 - Broad Institute, virtual, February 2024.
 - Two Sigma PhD Fellowship Finalist Reception, virtual, February 2024.
- *Provable Guarantees for Nonlinear Feature Learning in Three Layer Neural Networks.*
 - NDSEG Fellows Conference, San Antonio, July 2023.
- *Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability.*
 - Alg-ML seminar, Princeton University, December 2022.
- *Identifying good directions to escape the NTK regime and efficiently learn low-degree plus sparse polynomials.*
 - Joan Bruna group meeting, NYU, June 2022.

WORK EXPERIENCE

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| Flatiron Institute CCM – Summer Research Associate
<i>New York, NY</i> | June 2024 – August 2024 |
| <ul style="list-style-type: none"> • Studied a theoretical model for factual recall in large language models. Advised by Dr. Alberto Bietti. | |
| DeepMind – Research Engineering Intern
<i>London, United Kingdom</i> | June 2019 – August 2019 |
| <ul style="list-style-type: none"> • Researched and developed a novel general-purpose optimization algorithm for training deep neural networks. | |
| Asana – Software Engineering Intern
<i>San Francisco, CA</i> | June 2018 – August 2018 |

TEACHING EXPERIENCE

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| Teaching Assistant – Theoretical Machine Learning (Princeton COS 511) | September 2024 – December 2024 |
| Teaching Assistant – Algorithms for Inference (MIT 6.438) | September 2020 – December 2020 |
| Teaching Assistant – Design and Analysis of Algorithms (MIT 6.046) | September 2019 – May 2020 |
| Mathematics Coach – MIT Global Teaching Labs & MISE Foundation
<i>Accra, Ghana</i> | January 2020 |
| <ul style="list-style-type: none"> • Coached team of students who represented Ghana at the 2020 International Mathematical Olympiad. • Taught competition mathematics & problem solving skills at 10+ high schools and middle schools in Accra. | |

ACADEMIC SERVICE

- Reviewing:** ICML: 2022-2025; NeurIPS: 2022-2025 (2x **Outstanding Reviewer Award**); ICLR: 2025; COLT: 2025; JMLR; SIMODS; TMLR
- Mentoring:**
- Zihao Wang (Peking University undergrad), 2023.
- Leadership:**
- Princeton Alg-ML Seminar, co-organizer (2023-24 academic year).

PROGRAMMING LANGUAGES/FRAMEWORKS

- Python, JAX, PyTorch, TensorFlow, numpy.

GRADUATE COURSEWORK

- Princeton:** Optimization for Machine Learning, Deep Learning Theory, Foundations of Reinforcement Learning, Mathematical Understanding of Deep Learning, Theoretical Machine Learning, Probability in High Dimension, Statistical Theory and Methods, Theory of Deep Weakly Supervised Learning.
- MIT:** Theoretical Foundations for Deep Learning, Mathematical Statistics: A Non-Asymptotic Approach, Graph Theory and Additive Combinatorics, Statistical Learning Theory, Bayesian Modeling and Inference, Randomized Algorithms, Theory of Probability, Inference and Information, Machine Learning, Advanced Algorithms.